

Q1

Determine the time complexity: $O(n)$

```
for(int i=0;i<n;i++)  
{  
    cout<<i;  
}
```

Q2

Determine the time complexity: $O(n^2)$

```
for(int i=0;i<n;i++)  
{  
    for(int j=0;j<n;j++)  
    {  
        cout<<"*";  
    }  
}
```

Q3

Determine the time complexity: $O(n^2)$

```
for(int i=0;i<n;i++)  
{  
    for(int j=0;j<i;j++)  
    {  
        cout<<"*";  
    }  
}
```

```
}  
}
```

Q4

Determine the time complexity: $O(n \log n)$

```
for(int i=1;i<n;i*=2)  
{  
    cout<<i;  
}
```

Q5

Determine the time complexity: $O(n \log n)$

```
for(int i=n;i>1;i/=2)  
{  
    cout<<i;  
}
```

Q6

Determine the time complexity: $O(n \log n)$

```
for(int i=1;i<=n;i++)  
{  
    for(int j=1;j<=log(n);j++)  
    {  
        cout<<"*";  
    }  
}
```

```
}
```

Q7

Determine the time complexity: $O(n)$

```
while(n>0)
```

```
{
```

```
    n--;
```

```
}
```

Q8

Determine the time complexity: $O(\log n)$

```
while(n>1)
```

```
{
```

```
    n/=2;
```

```
}
```

Q9

Determine the time complexity: $O(n)$

```
for(int i=1;i<=n;i++)
```

```
{
```

```
}
```

```
for(int j=1;j<=n;j++)
```

```
{
```

```
}
```

Q10

Determine the time complexity: $O(n^2)$

```
for(int i=1;i<=n;i++)  
{  
}  
for(int j=1;j<=n*n;j++)  
{  
}
```

Q11

Determine the time complexity: $O(n^3)$

```
for(int i=1;i<=n;i++)  
{  
    for(int j=1;j<=n*n;j++)  
    {  
    }  
}
```

Q12

Determine the time complexity:

```
for(int i=1;i<=n;i*=3)  $O(n \log n)$ - Here log is base 3  
{  
}
```

Q13

Determine the time complexity: $O(n^3)$

```
for(int i=1;i<=n;i++)  
{  
    for(int j=1;j<=i*i;j++)  
    {  
    }  
}
```

Q14

Determine the time complexity: $O(n)$

```
void fun(int n)  
{  
    if(n==0)  
        return;  
  
    fun(n-1);  
}
```

Q15

Determine the time complexity: $O(\log n)$

```
void fun(int n)  
{  
    if(n<=1)  
        return;
```

```
    fun(n/2);  
}
```

Q16

Determine the time complexity: $O(2^n)$

```
void fun(int n)  
{  
    if(n==0)  
        return;
```

```
    fun(n-1);  
    fun(n-1);  
}
```

Q17

Determine the time complexity: $O(2^n)$

```
int fib(int n)  
{  
    if(n<=1)  
        return n;
```

```
    return fib(n-1)+fib(n-2);  
}
```

Q18

Determine the time complexity: $O(n)$

```
int fact(int n)
{
    if(n<=1)
        return 1;

    return n*fact(n-1);
}
```

Q19

Determine the time complexity: $O(n)$

```
void fun(int n)
{
    if(n<=0)
        return;

    fun(n-1);

    cout<<n;
}
```

Q20

Determine the time complexity: $O(\log n)$

```
void fun(int n)
{
    if(n<=1)
```

```
    return;  
  
    fun(n/2);  
  
    cout<<n;  
}
```

Q21

Solve the recurrence relation:

$$T(n) = T(n-1) + 1$$

Q22

Solve the recurrence relation:

$$T(n) = T(n-1) + n$$

Q23

Solve the recurrence relation:

$$T(n) = T(n/2) + 1$$

Q24

Solve the recurrence relation:

$$T(n) = 2T(n-1) + 1$$

Q25

Solve the recurrence relation:

$$T(n) = T(n/2) + n$$

Bonus GATE-Level Questions

Q26

TC : $O(n \log n)$

```
for(int i=1;i<=n;i++)  
{  
    for(int j=1;j<=i;j*=2)  
    {  
    }  
}
```

Determine the complexity.

Q27

TC : $O(n^2 \log n)$

```
for(int i=1;i<=n;i*=2)  
{  
    for(int j=1;j<=n;j++)  
    {  
    }  
}
```

Determine the complexity.

Q28

TC : $O(n^2)$

```
for(int i=n;i>=1;i--)  
{  
    for(int j=i;j>=1;j--)
```

```
{  
}  
}
```

Determine the complexity.

Q29 TC : $O(n^2 \log n)$

```
for(int i=1;i<=n;i++)  
{  
    for(int j=1;j<=n;j*=2)  
    {  
    }  
}
```

Determine the complexity.

Q30 TC : $O(n)$

```
void fun(int n)  
{  
    if(n<=1)  
        return;  
  
    fun(n/2);  
    fun(n/2);  
}
```

Determine the complexity.